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Questions

Q1.

Parkinson's disease occurs when dopamine-producing neurones die.

(a) The most obvious motor symptoms of Parkinson's disease include muscle tremors and muscle rigidity.

These motor symptoms are linked to the loss of dopamine-producing neurones in the brain.

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(i) One drug used to treat Parkinson's disease is L-DOPA.

Which of the following statements explains why L-DOPA can be used to treat Parkinson's disease?

(1)

☒ **A** L-DOPA acts on a different post-synaptic receptor to dopamine

☒ **B** L-DOPA crosses the blood-brain barrier and is then converted to dopamine in the brain

☒ **C** L-DOPA is converted to dopamine in the blood and then crosses the blood-brain barrier

☒ **D** L-DOPA is identical to dopamine

(ii) Explain why the release of reduced quantities of dopamine by presynaptic neurones could result in motor symptoms.

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(b) It has been suggested that in Parkinson's disease dopamine-producing neurones die because of a failure to release dopamine.

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Suggest how the release of dopamine from presynaptic neurones could be inhibited.

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(c) Describe how microarrays and bioinformatics could be used to investigate the genetic basis of Parkinson's disease.

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(Total for question = 10 marks)

Q2.

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One cause of blindness is a mutation in the DNA coding for the enzyme RPE65.

The enzyme RPE65 converts trans-retinal to cis-retinal.

(a) Explain why mutations in the RPE65 gene can result in blindness.

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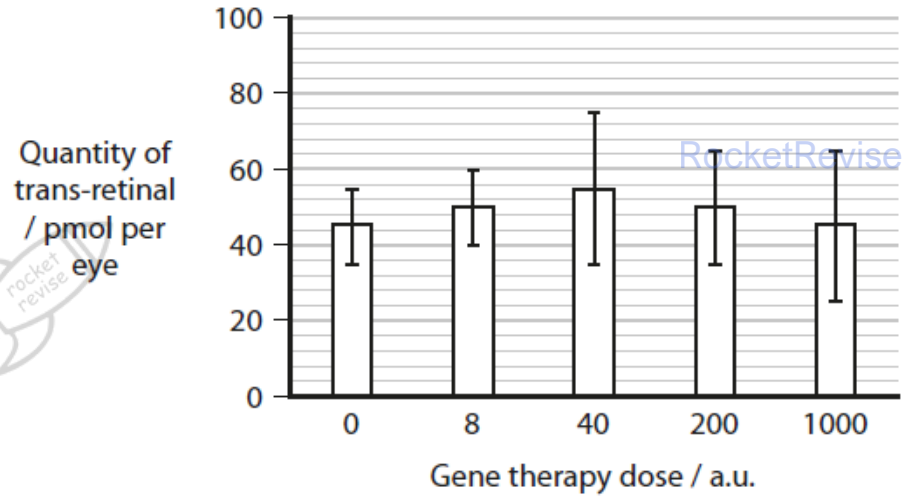
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(b) Scientists are investigating the use of gene therapy to treat blindness caused by mutations in the RPE65 gene.

In one investigation, different gene therapy doses containing functioning RPE65 genes were compared.

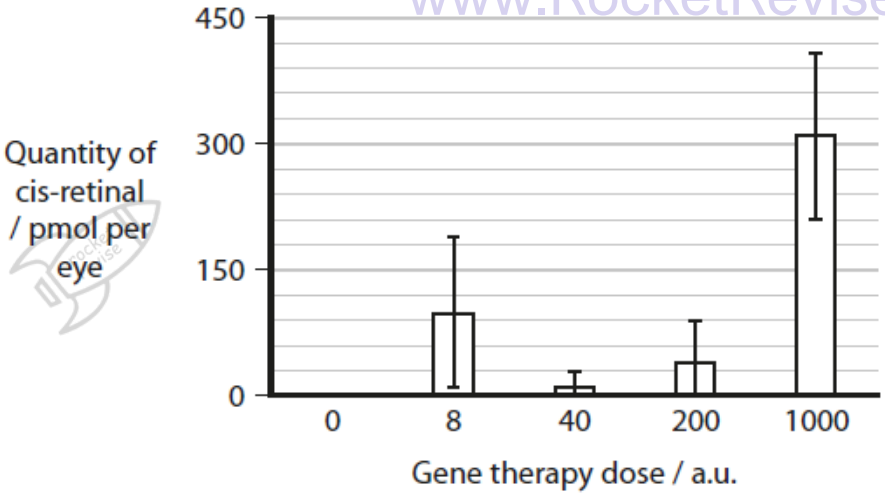
Two months after treatment, the trans-retinal and cis-retinal in each eye and the ability to follow a path were measured.

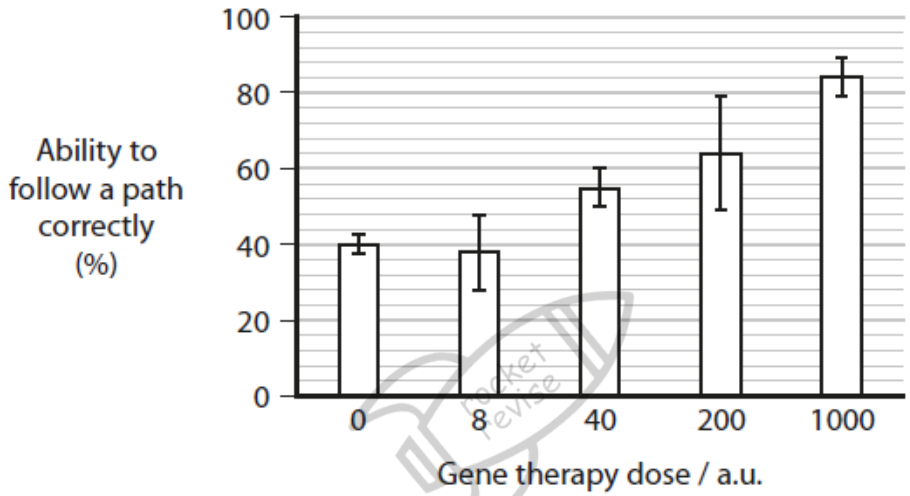
The graphs show the results of this investigation.



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Describe the conclusions that can be made from this investigation.

Use the information in the graphs to support your answer.

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(c) Mutations in other genes can result in blindness.

Describe how these genes could be identified.

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(Total for question = 10 marks)

Q3.

Vitamin A is an essential component of the human diet.

Rice is a staple food in parts of the world where a lot of people have vitamin A deficiency.

Rice contains no β-carotene or vitamin A.

Cells in the human intestine can convert β-carotene from the diet to vitamin A.

Some soil bacteria can produce large quantities of β-carotene.

(a) Describe how genetic engineering could be used to produce rice that contains β-carotene.

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(b) (i) Describe the possible risks of growing genetically modified rice plants.

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(ii) Suggest how one of these risks could be reduced.

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(1)

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(Total for question = 8 marks)

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Q4.

The scientific article you have studied is adapted from *Rethinking Alzheimer's* in *New Scientist* (February 2019).

Use the information from the scientific article and your own knowledge to answer the following questions.

(a) Memory and cognitive function are two processes that occur in the brain (paragraphs 2 and 13).

Different parts of the brain are responsible for specific functions.

Complete the table showing which parts of the brain are responsible for different functions.

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(2)

Part of brain	Function
cerebellum	
	interpretation of information from the retina
medulla oblongata	
	secretion of ADH

(b) Some forms of dementia can be diagnosed using brain scans.

Compare and contrast the use of functional magnetic resonance imaging (fMRI) and computed tomography (CT) scans to identify structures like plaques in the brain (paragraph 6).

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(c) Suggest how the amyloid protein may prevent bacterial infection in the brain (paragraphs 7 and 22).

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(d) Explain how the bacterium *Porphyromonas gingivalis* that causes gum disease may invade the brain and cause inflammation in brain regions affected by Alzheimer's disease (paragraphs 10, 20 and 21).

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(e) Suggest how gingipains may have a degrading effect on the memory of a person suffering from Alzheimer's disease (paragraph 12).

(3)

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(f) The team found genetic material from *P. gingivalis* in the cerebral cortex (paragraph 13).

Describe how the team could show that this genetic material is active.

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(Total for question = 20 marks)

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Q5.

The scientific document you have studied is adapted from an article in Science Direct: *Assessment of food toxicology* by Alexander Gosslau.

Use the information from the scientific article and your own knowledge to answer the following questions.

(a) Explain how toxicants in food could cause Parkinson's disease (paragraph 3).

(3)

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(b) Suggest how the genes affected by toxicants can be identified (paragraph 4).

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(c) Explain how scientists could determine the high degree of homology of genes between humans and zebrafish (paragraph 5).

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(d) Suggest how electrophilic species (ES) of free radicals could be "involved in the modulation of gene expression" (paragraph 6).

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(e) Suggest how ES can damage mtDNA (paragraph 7).

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(f) In the MTT assay, reduced NAD converts an MTT solution from yellow to purple.

Describe how the MTT assay could be used to show which of two drugs is more toxic to cells (paragraph 10).

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(g) Explain the advantages of reprogramming ordinary somatic cells to behave like embryonic stem cells for use in stem cell-based assays of toxicants (paragraph 11).

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(Total for question = 20 marks)

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Q6.

The scientific document you have studied is adapted from an article '*How we perceive the world*' *by Julius D. and Patapoutian A*. in 2021.

Use the information from the scientific document and your own knowledge to answer the following questions.

(a) Describe how light is detected by the rod cells in the eye (paragraph 1).

(4)

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(b) Suggest how the genes expressed in a sensory neurone could be identified (paragraph 4).

(3)

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(c) Explain how capsaicin could activate nerve cells causing a pain sensation (paragraph 4).

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(d) Explain how changes in ion channel protein can result in nerve impulses in the nervous system (paragraph 5).

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(e) Explain how mechanical stimuli, e.g. pressure, are converted into electrical signals in cells (paragraphs 5 and 7).

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(f) Explain what is meant by the phrase 'Based on its similarity to Piezo 1, a second ion channel was found (Piezo 2)' (paragraph 7 and Figure 3).

(2)

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(g) Suggest the role of TRPV1 in maintaining core body temperature (paragraph 9).

(2)

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(Total for question = 20 marks)

Q7.

Animals and plants can be genetically modified.

(a) Recombinant human erythropoietin (rHE) is a protein drug that can be given to patients with low haemoglobin concentrations.

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(i) Sheep have been genetically modified to produce rHE in their milk.

Explain how a sheep could be genetically modified to produce rHE in its milk.

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(ii) In a clinical study, 4382 patients were treated with rHE. They showed a mean increase in blood haemoglobin from 80 g dm^{-3} to 142 g dm^{-3}.

Calculate the percentage increase in the mean blood haemoglobin concentration as a result of rHE treatment.

(1)

Answer ..... %

(iii) Some patients receiving rHE develop anti-rHE antibodies.

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Describe how these antibodies may be produced.

(4)

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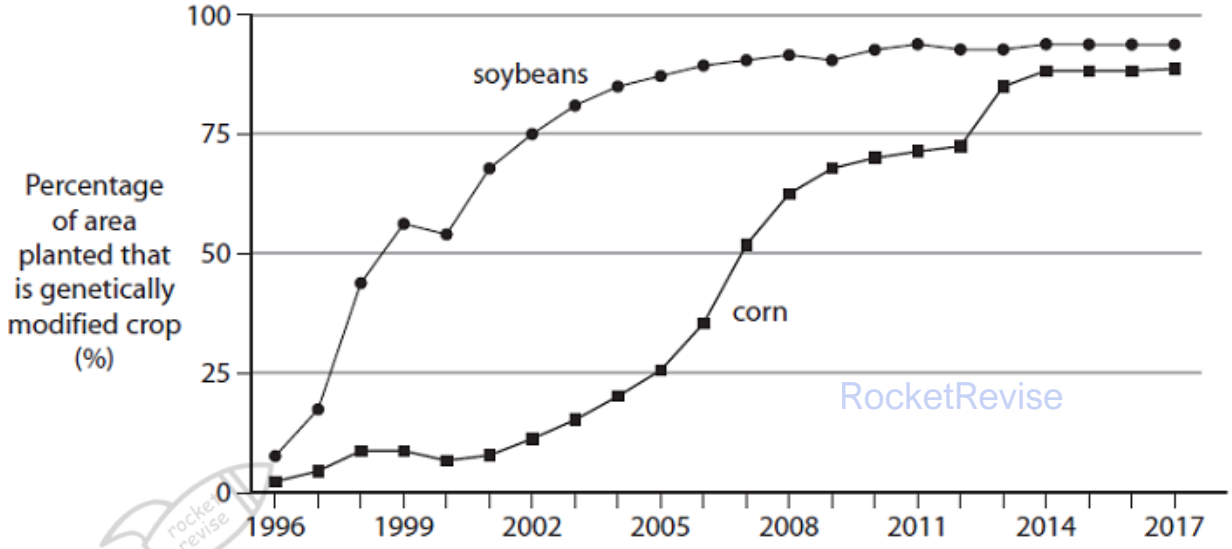
(b) Advances in genetic engineering have allowed for precise control of the genetic changes introduced into plants.

The table describes some examples of the use of genetic modification in plants.

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Trait	Plant	Details
herbicide tolerance	soybean	glycophosphate herbicide tolerance using a gene from the bacterium *A. tumefaciens*
insect resistance	corn	resistance to European corn borer using a gene from *B. thuringiensis*

The graph shows the increase in the use of genetically modified crops in the United States from 1996 to 2017.



Discuss the risks and benefits of genetic modification in valuable agricultural plants.

Use all the data and your own knowledge to support your answer.

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(Total for question = 15 marks)

Q8.

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Parkinson's disease is caused by a loss of neurones in part of the brain called the substantia nigra.

Co-beneldopa is used to treat Parkinson's disease.

Co-beneldopa is a mixture of L-dopa and a drug that prevents the conversion of L-dopa into dopamine in the blood.

(a) Explain why using Co-beneldopa will be more effective than using L-dopa alone to treat Parkinson's disease.

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(b) In an investigation, tomato plants were genetically modified to convert the amino acid tyrosine into L-dopa.

Three genes were used in this investigation:

- a control gene that is not involved in the production of L-dopa
- MYB12, a plant gene that produces a transcription factor
- CYP76, a gene found in beetroot that codes for an enzyme that converts tyrosine into L-dopa.

The table shows the concentration of L-dopa in the tomato fruit from genetically modified plants.

Genes used to modify the tomato plants	Concentration of L-dopa in the tomato fruit / mg 100 g^{-1}
none	1.2
control gene	1.2
MYB12	1.5
CYP76	10.2
CYP76 and MYB12	14.6

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(i) Explain the effects of these three genes in this investigation.

(4)

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(ii) Describe how tomato plants could be genetically modified to make the enzyme that converts tyrosine to L-dopa.

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(Total for question = 10 marks)

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Q9.

Alpha-1 antitrypsin (AAT) deficiency is an inherited disease that can affect different organ systems.

Alpha-1 antitrypsin (AAT) is a protein produced in the liver.

(a) Investigation of the breathing of a patient with AAT deficiency found that the:

- resting tidal volume (V_T) was 500 cm^3

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- dead space (DS) was 150 cm^3
- resting respiratory rate (RR) was 12 breaths per minute.

Calculate the volume of air that enters the alveoli (A_V) **each hour**.

$$A_V = (V_T - DS) \times RR$$

Give your answer to 2 significant figures with appropriate units.

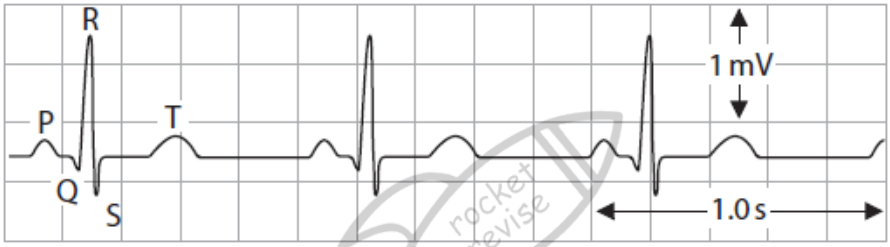
(3)

Answer .....

(b) Activity of the heart was investigated in the same patient.

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The trace shows a normal electrocardiogram (ECG).



(i) Name the stage of the cardiac cycle at T on this ECG.

(1)

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(ii) The ECG of the patient with AAT is shown in the trace.



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Calculate the mean heart rate shown in this ECG trace.

(1)

Answer .....

(iii) Compare and contrast the ECG trace of this patient with the normal ECG trace.

(3)

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(c) One potential treatment for AAT deficiency uses adenoviruses to genetically modify cells in an affected person.

Adenoviruses contain double-stranded DNA.

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Explain how adenoviruses can be used to treat AAT deficiency in a person.

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(Total for question = 12 marks)

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Q10.

The scientific document you have studied is adapted from the article '*Brain aging and neurodegeneration: from a mitochondrial point of view*' by Amandine Grimm and Anne Eckert.

Use the information from the article and your own knowledge to answer the following questions.

(a) Suggest why there is a relationship between the ageing process and neurodegeneration (paragraphs 1 and 2).

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(3)

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(b) Describe how energy released from glucose is used in oxidative phosphorylation (paragraph 2).

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(c) Explain how free radicals (reactive oxygen species, ROS) can cause damage to mitochondrial DNA
(paragraphs 2 and 4).

(3)

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(d) "The brain is a remarkable organ composed by highly differentiated cells... with different morphology, according to their role and their localization" (paragraph 6).

Describe how this differentiation can occur.

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(e) Explain what is meant by the phrase 'mitochondrial proteome' (paragraph 7).

(2)

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(f) Explain why synaptic mitochondria are more susceptible to ageing than the mitochondria found in other parts of a neurone (paragraph 9).

(2)

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(g) Explain why the mitochondrial genome is essential for oxidative phosphorylation (paragraph 10).

(2)

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(h) Suggest how damaged mitochondria are removed by 'autophagy' (paragraph 11).

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(Total for question = 20 marks)

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Q11.

Dopamine, serotonin and acetylcholine are neurotransmitters in the human nervous system.

(a) Describe how a neurotransmitter allows an impulse to be transferred to an adjacent neurone.

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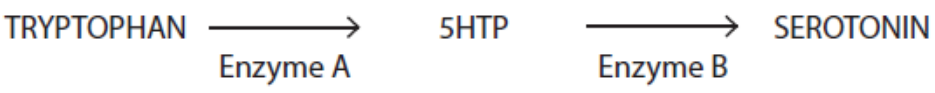
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(b) The diagram shows how serotonin is produced from tryptophan.



Enzymes A and B are found in the Madagascar periwinkle.

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Medical serotonin can be produced by genetically engineered *E.coli* bacteria.

Describe how *E.coli* could be genetically engineered using the Madagascar periwinkle to produce serotonin.

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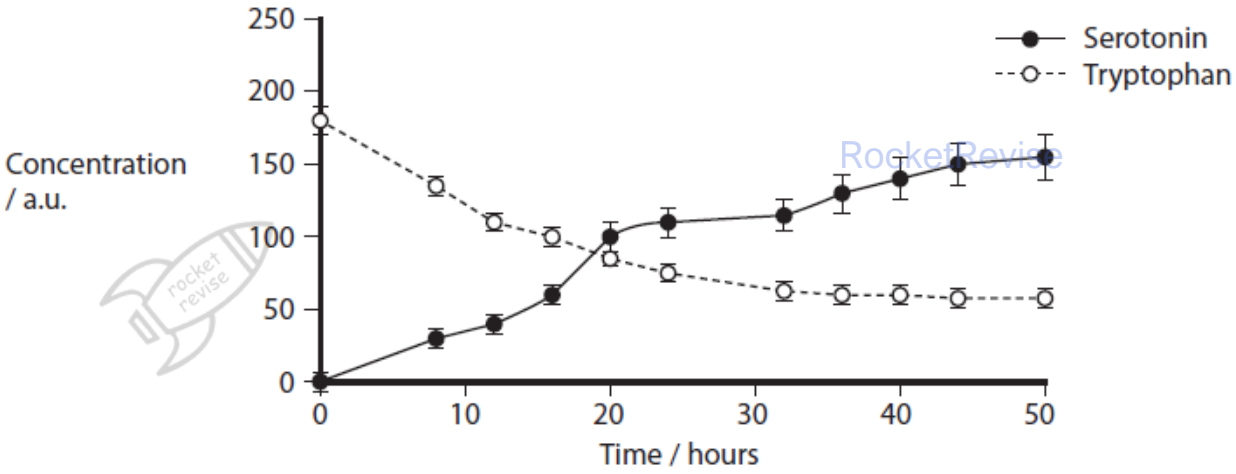
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(c) The genetically engineered *E. coli* was grown in a liquid culture.

The concentrations of tryptophan and serotonin in the culture were measured.

The graph shows the results.



Discuss the changes shown in the graph.

Use the information in the question to support your answer.

(4)

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(Total for question = 12 marks)

Q12.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The image shows a section of a recombinant DNA molecule.

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(a) (i) Which enzyme is used in the production of recombinant DNA?

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(1)

☒ **A** acetylcholinesterase

☒ **B** amylase

☒ **C** restriction endonuclease

☒ **D** RNA polymerase

(ii) DNA is present in bacterial cells.

Where is DNA found in a bacterial cell?

(1)

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☒ **A** nucleus only

☒ **B** plasmid only

☒ **C** plasmid and cytoplasm

☒ **D** plasmid and nucleus

(iii) Which bond joins nucleotides together in a **single** strand of DNA?

(1)

☒ **A** hydrogen

☒ **B** ionic

☒ **C** peptide

☒ **D** phosphodiester

(b) Describe how genes can be switched **off** by transcription factors.

(3)

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(Total for question = 6 marks)

Q13.

The scientific document you have studied is adapted from the article '*Microbiota-gut-brain axis and the central nervous system*' by Xiqun Zhu, Yong Han, Jing Du, Renzhong Liu, Ketao Jin and Wei Yi in *Oncotarget* (2017).

Use the information from the article and your own knowledge to answer the following questions.

(a) "Gut microorganisms play an important role... affecting human metabolic functions by decomposing the complex polysaccharides in food".

Explain the role of gut microorganisms in decomposing complex polysaccharides (paragraph 2).

(3)

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(b) Describe how the vagus nerve is involved in the control of heart rate (paragraph 6).

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(c) "Microorganisms can also enhance the anti-tumor immune effect of drugs by promoting T cell accumulation and transformation."

Explain how microorganisms promote the accumulation of mature T-killer cells that enhance the anti-tumour effects of drugs (paragraph 7).

(3)

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(d) Describe how fMRI scans could be used to show the activity of the regions of the brain controlling memory and sensation when probiotics are consumed (paragraph 11).

(2)

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(e) Suggest how changes in diet could affect the composition of the gut flora causing imbalances in the naturally occurring chemicals in the brain (paragraph 10).

(3)

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(f) "Multiple sclerosis (MS) is a... demyelinating disease of the nervous system" (paragraph 14).

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Symptoms of MS include poor coordination and loss of vision.

Explain how demyelination could result in these symptoms.

(2)

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(g) Gut microorganisms produce LPS (paragraphs 15 and 17).

White blood cells have receptors for LPS on their cell surface.

Describe the techniques that could be used to identify the LPS receptor gene found in white blood cells.

(3)

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(Total for question = 20 marks)

Q14.

Some groups of living organisms produce neurotoxins.

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Neurotoxins act at synapses, preventing the generation of nerve impulses.

Some scientists have been studying neurotoxins using a combination of computer science, statistics and mathematics.

(a) What is the name of a study that combines computer science, statistics and mathematics?

(1)

☒ **A** bioinformatics

☒ **B** computed tomography

☒ **C** epigenetics

☒ **D** forensic entomology

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(b) Some neurotoxins act in these ways:

- inhibiting the release of acetylcholine
- blocking of ion channels
- blocking of acetylcholine receptors.

Explain how these neurotoxins prevent the generation of nerve impulses.

(3)

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(c) The table shows the mean composition of three amino acids, in neurotoxins and non-toxins.

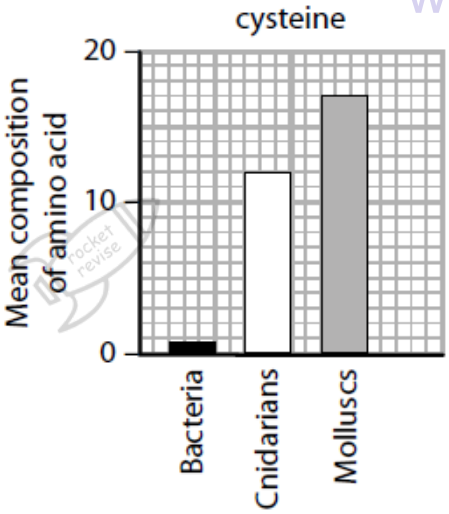
Type of amino acid	Mean composition of amino acid in neurotoxins	Mean composition of amino acid in non-toxins
Cysteine	13.69	1.95
Glutamine	2.35	4.16
Leucine	5.53	10.17

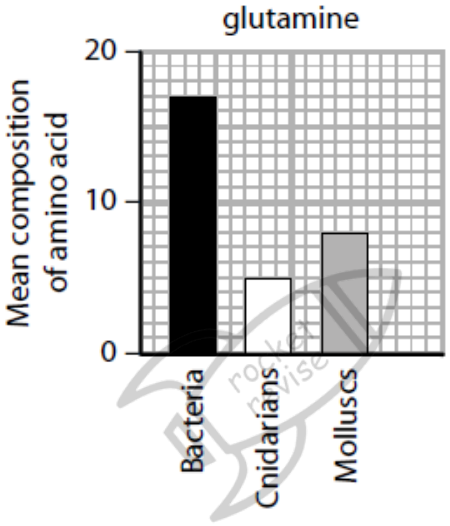
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The graphs show the mean composition of these amino acids in neurotoxins produced by three groups of organisms: bacteria, cnidarians and molluscs.

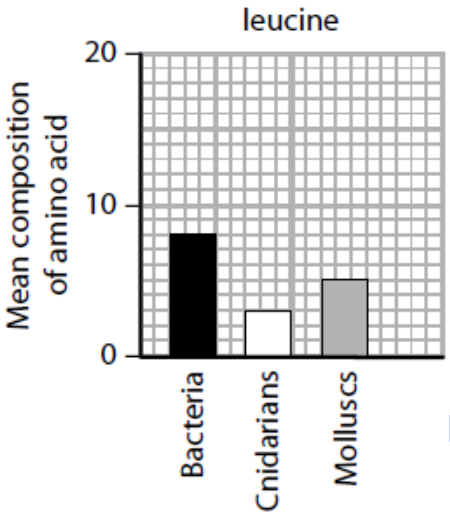
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*(i) Comment on the amino acid composition of these neurotoxins. Use the information in the table and in the graphs to support your answer.

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(ii) Explain why the replacement of one amino acid in a neurotoxin could decrease its toxicity.

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(Total for question = 12 marks)

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Q15.

The scientific document you have studied is adapted from an article in nature.com: *Microgravity elicits reproducible alterations in cytoskeletal and metabolic gene and protein expression in space-flown Caenorhabditis elegans*.

Use the information from the scientific document and your own knowledge to answer the following questions.

(a) Suggest why the authors of this article have named the species of nematode used in this study (paragraph 2).

(1)

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(b) State **one** ethical issue of using nematodes in this study.

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(c) Describe how microgravity can cause changes in gene expression in these nematodes (paragraph 4).

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(d) Suggest why there was a lower fat accumulation and shorter body length in nematodes grown in microgravity (paragraph 4).

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(e) Explain why there were reduced levels of a number of metabolic proteins in nematodes grown in microgravity (paragraph 6).

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(f) Explain why microarrays could be used to show that mitochondrial electron transport genes are downregulated, whereas the sirtuin gene was upregulated (paragraph 7).

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(g) Explain why Sudan Black was used in this study (paragraph 8).

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(h) Comment on the reliability of the data presented in Table 1 (paragraph 9).

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(Total for question = 20 marks)

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Q16.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Small variations in DNA sequences and activation of genes can explain why individuals respond differently to diseases and the drugs used to treat them.

(a) Describe how active genes could be identified.

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(b) Epigenetic modification can have an effect on the activity of genes in a cell.

(i) Which is the role of histones in the cell?

(1)

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☒ **A** act as a transcription factor

☒ **B** form pores in the nuclear membrane

☒ **C** give structural support for DNA molecules

☒ **D** produce RNA polymerase

(ii) Which is the process that regulates gene expression?

(1)

☒ **A** crossing over

☒ **B** epigenetic changes

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☒ **C** post-transcriptional changes

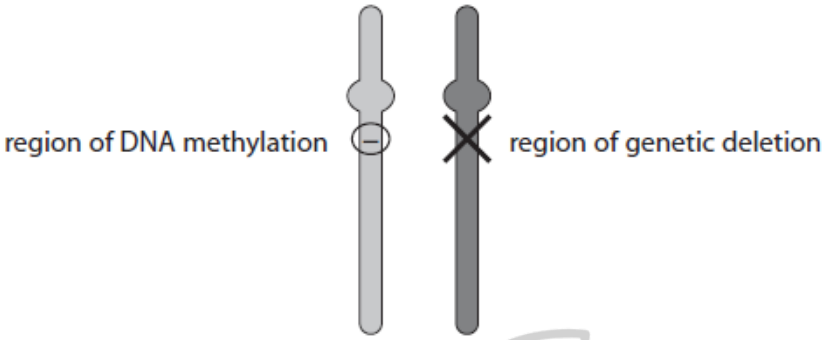
☒ **D** post-translational changes

(c) Prader–Willi syndrome is an inherited genetic disorder that involves changes on chromosome 15.

The chromosome inherited from the father has genetic deletions.

The chromosome inherited from the mother has DNA methylated bases.

The diagram shows these chromosomes for an individual with Prader–Willi syndrome.



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(i) Explain why these changes in the chromosome of this individual would cause Prader–Willi syndrome.

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(ii) The symptoms of Prader–Willi syndrome can be treated using human growth hormone (HGH).

This hormone is a protein produced by the pituitary gland.

Describe how *Escherichia coli* bacteria can be used to produce human growth hormone.

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(Total for question = 11 marks)

Q17.

The scientific article you have studied is adapted from articles about Alzheimer's disease published in *The Conversation*: August 31, 2023 and November 13, 2022

(a) Describe the possible role of hypertension (high blood pressure) in causing dementia (Paragraphs 2 and 19).

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(2)

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(b) Suggest why the protein beta-amyloid from the fatty membranes surrounding nerve cells 'is

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chemically sticky, gradually clumping together' (Paragraph 7).

Use your knowledge of protein structure to support your answer.

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(c) Describe the information that would be obtained from using a combined MRI and PET scan of the
brain (Paragraph 13).

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(d) State **one** reason why it might be considered unethical to tell people they are at risk of developing cognitive impairment (Paragraph 15).

(1)

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(e) Explain why a 'falling blood pressure' might result in poor brain health (Paragraph 20).

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(f) Some genes may be linked to an increase in the risk of dementia.

Describe how scientists could identify the link between these genes and the risk of dementia (Paragraph 28).

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(g) Describe what is meant by the term 'monoclonal antibody' (Paragraph 31).

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(h) Give **one** reason why a placebo was used in the donanemab trial (Paragraph 33).

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(i) Swelling is a symptom of inflammation.

Explain why lecanemab may result in 'brain swelling' (Paragraph 35).

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(j) Successfully clearing amyloid from the brain had only a small effect.

Suggest **one** reason why researchers are suggesting the need to start using the drugs even 'earlier in the disease' (Paragraphs 44 and 45).

(1)

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(Total for question = 20 marks)

Q18.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Some diseases, and certain cancers, are thought to be caused by a lack of specific neurotransmitters.

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This means they can be treated using drugs which increase the concentration of these neurotransmitters.

Some of these drugs are manufactured using genetically modified (GM) organisms.

(a) Which is a description of a GM organism?

(1)

☒ **A** an amplified sequence of nucleic acid

☒ **B** an organism containing genetic material from another species

☒ **C** a virus containing DNA

☒ **D** DNA that contains a mutation

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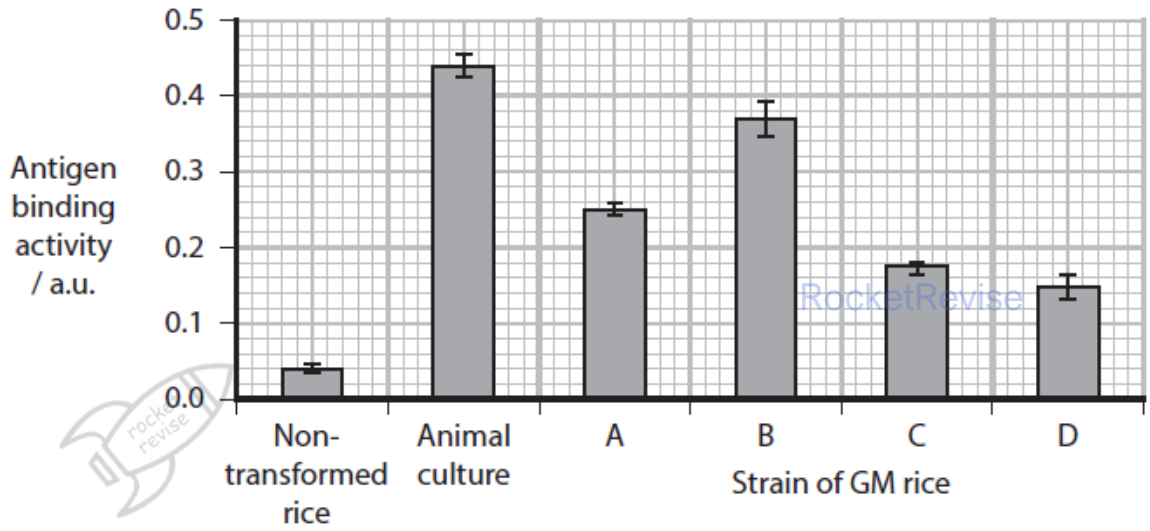
(b) Bevacizumab (BVZ) is an antibody used to treat brain cancer patients.

This antibody is produced using animal cell cultures.

A new method of producing BVZ has been developed using genetically modified (GM) rice.

The graph compares the antigen binding activity of a BVZ antibody produced using:

- a control using non-transformed rice
- animal cell culture
- four strains of GM rice (A, B, C and D).



(i) Describe **three** conclusions that can be made from these results.

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(ii) Explain how BVZ antibodies can cause the destruction of cancer cells.

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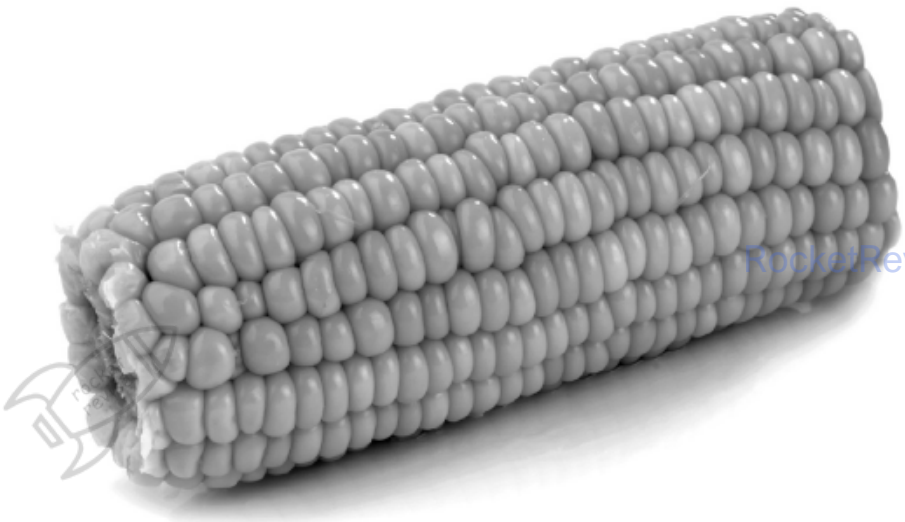
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*(c) Maize is a major food crop.

The photograph shows a maize cob with rows of grains.



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(Source: © art nick/Shutterstock)

Maize can be infected by a fungus.

This fungus produces a toxin that affects plant growth and is poisonous to animals.

This fungus can be genetically modified (GM).

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In an investigation, maize plants were infected with normal fungus or infected with GM fungus.

The table shows the mean and standard deviation for growth measurements of the maize and the concentration of toxin.

Crop infected with	Plant height / cm	Number of grains per cob	Concentration of toxin / a.u.
Normal fungus	285.0 ± 60	113 ± 2	530 ± 100
GM fungus	320.0 ± 33	117 ± 2	270 ± 35

A computer model was used to predict the effect of global warming on the percentage of maize crop infected with this fungus.

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The results are shown in the table.

Temperature	Percentage of maize crop predicted to be infected with this fungus (%)
Current temperature	38
Current temperature + 2°C	73
Current temperature + 5°C	95

Assess the risks and benefits of genetically modifying the fungus for future food production in a globally warming world.

Use the information given in the tables together with your own knowledge to support your answer.

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(Total for question = 12 marks)

Q19.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The nervous system of an organism enables it to respond to a stimulus.

(a) Nociceptors and thermoreceptors are receptors that are sensitive to changes in temperature.

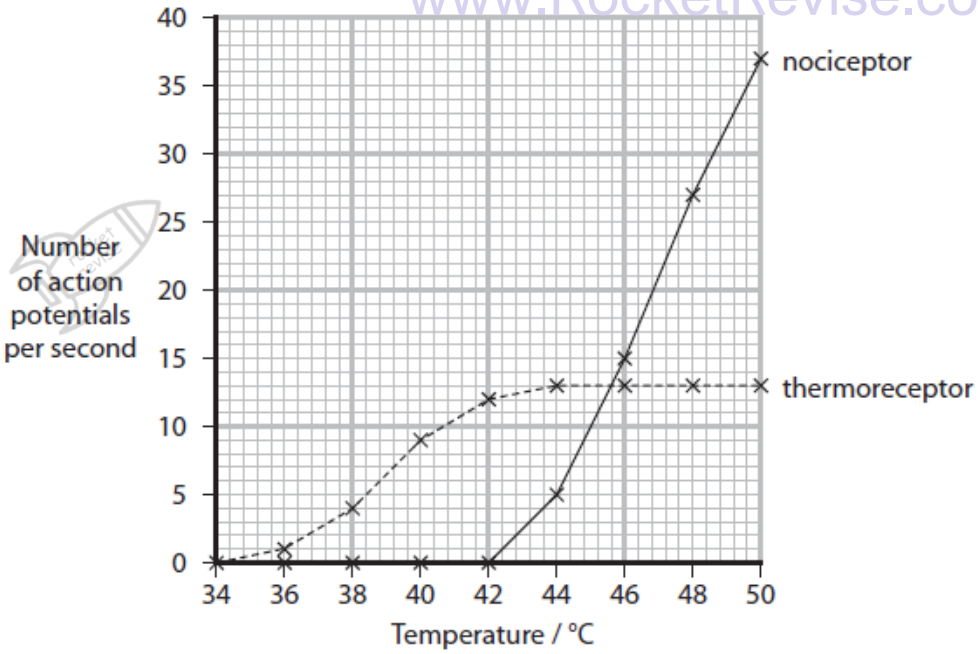
An investigation compared the effects of temperature on the activity of thermoreceptors and nociceptors in the skin of volunteers.

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The graph shows the number of action potentials produced when each type of receptor was stimulated at different temperatures.

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(i) Which is the ratio of the action potentials produced at 48 °C by nociceptors compared with thermoreceptors?

(1)

☒ **A** 0.40 : 1

☒ **B** 2.0 : 1

☒ **C** 2.07 : 1

☒ **D** 2.1 : 1

(ii) Nociceptors are better than thermoreceptors at preventing serious damage to tissues at high temperatures.

Comment on the extent to which the results of this investigation support this statement.

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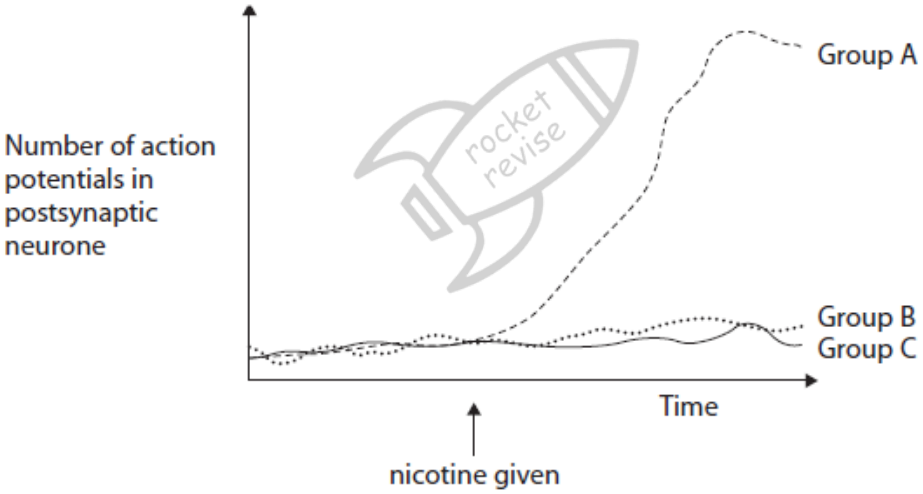
(b) Scientists used three groups of mice (A, B and C) to investigate the effect of nicotine on the transmission of nerve impulses.

- Group A were given nicotine
- Group B were not given nicotine
- Group C were given nicotine but were deficient in the gene that produces the acetylcholine receptor on the postsynaptic membrane.

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The mice in groups A and C were given the same dose of nicotine (0.5 mg / kg) over the same time period.

The results are shown in the graph.



Explain the results of this investigation.

Use your own knowledge and the results in the graph to support your answer.

(4)

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(Total for question = 9 marks)

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Q20.

The scientific document you have studied is adapted from articles in *New Scientist* and the *Journal of Consumer Protection and Food Safety*.

Use the information from the scientific document and your own knowledge to answer the following questions.

(a) Explain how Ted Turlings and colleagues could have identified that the *BtPMaT1* gene found in the whitefly was originally a plant gene (paragraph 2).

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(b) Explain why the frequency of the *BtPMaT1* gene increases in the whitefly population (paragraphs 3 and 6).

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(c) Explain how populations of fluorescent red zebrafish could be produced for breeding (paragraphs 9 and 15).

(4)

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(d) Suggest the risks and benefits of genetically modifying zebrafish (paragraphs 9, 14 and 20).

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(e) Suggest how the skin cells 'doubled in size to cover the area of the damage' (paragraph 19).

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(f) Explain how a single-celled embryo of a zebrafish, injected with the colour producing gene, could develop into an adult with red skin cells (paragraphs 14, 15 and 17).

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(Total for question = 20 marks)

Q21.

Insulin is a peptide hormone produced by the pancreas.

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Insulin plays a crucial role in regulating blood glucose levels and maintaining homeostasis in the body.

(a) State what is meant by the term **homeostasis**.

(1)

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(b) The insulin concentration in blood is affected by many variables.

The effect of exercise on the concentration of insulin in the blood of trained and untrained individuals was investigated.

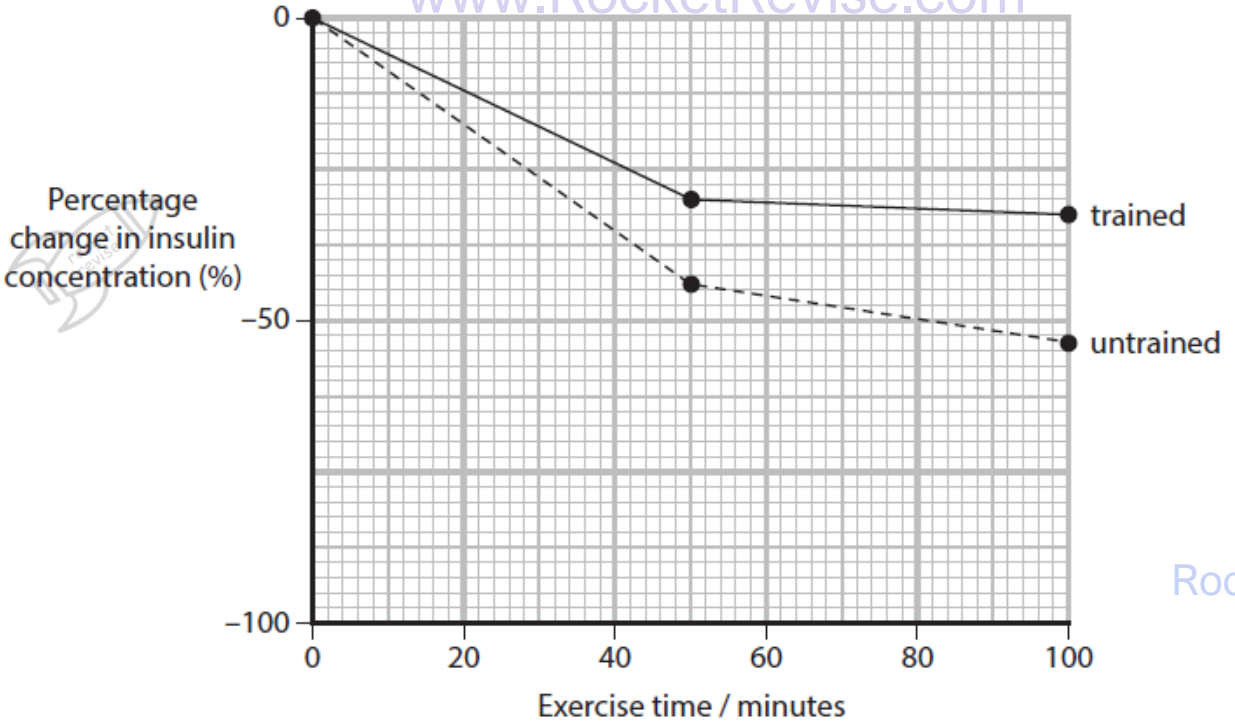
The graph shows the results of this investigation.

The graph compares the changes in insulin levels in an untrained individual and a trained individual.

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Compare and contrast the changes in concentration of insulin in the blood of these two individuals.

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(c) Diabetes is a disease that affects millions of people across the world.

Human insulin is a small molecule that consists of two polypeptide chains linked by disulfide bridges.

Insulin from animals can be used to treat diabetes.

Human insulin can now be synthesised using genetically modified bacteria.

(i) Describe how enzymes can be used to genetically modify the bacteria to produce human insulin.

(3)

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(ii) Suggest **two** advantages of producing **human** insulin using genetically modified bacteria compared with insulin produced from animals.

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(d) Insulin is a peptide hormone that can stimulate cell division.

In an investigation, cells were incubated in solutions containing different concentrations of insulin.

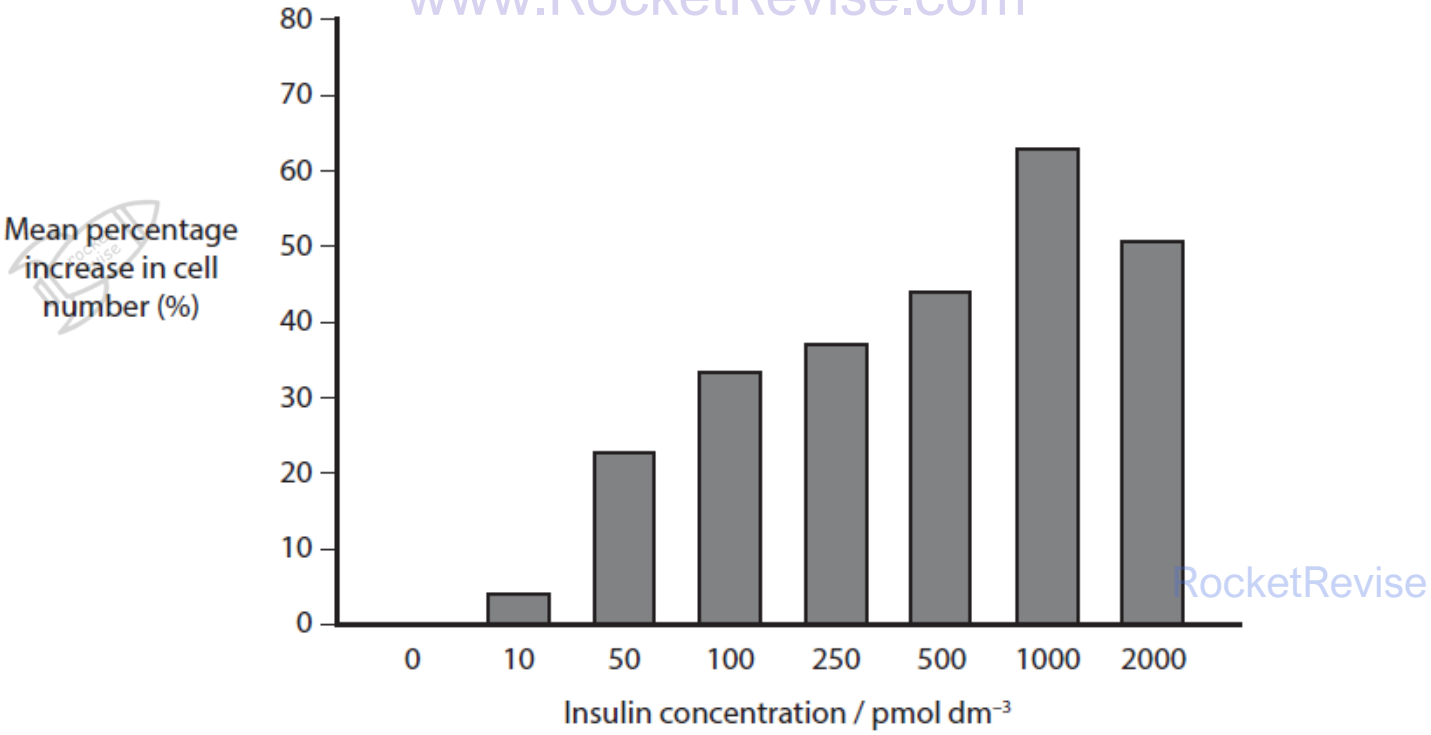
After 48 hours, the increase in cell number was measured.

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The results are shown in the graph.

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(i) Give **one** reason why there is no bar on the graph for an insulin concentration at 0 pmol dm^{-3}.

(1)

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(ii) Describe how the investigation was designed so that error bars could be produced.

(2)

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(iii) State how the error bars can be used to interpret the results of this investigation.

(1)

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(Total for question = 13 marks)

Q22.

The scientific document you have studied is adapted from the articles *Risk of heart failure may increase with age due to Y chromosome loss and Essence of man: Y size doesn't matter* in the New Scientist.

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Use the information from the scientific document and your own knowledge to answer the following questions.

(a) (i) Name the type of microscope that would be used to produce the image before paragraph 1.

(1)

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(ii) Calculate the ratio of the length of the X chromosome to the Y chromosome in this image.

Give your answer to **one** decimal place.

(2)

Answer ..... : 1

(b) Suggest why the loss of the Y chromosome may lead to higher rates of heart disease in men than women (paragraphs 4 and 5).

(2)

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(c) Describe the roles of nucleic acids and enzymes used in CRISPR gene editing to remove some of the Y chromosome genes (paragraph 5).

(2)

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(d) Describe how the cardiovascular health of participants in the UK Biobank study could be assessed (paragraph 6).

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(e) Explain how an antibody could block the effects of an inflammatory signalling molecule (ISM) (paragraphs 7 and 8).

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(f) Explain how environmental factors such as temperature could determine the sex of an animal (paragraph 14).

(2)

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(g) Calculate the percentage of genes that have been lost from the original Y chromosome (paragraph 15).

(1)

Answer .....%

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(h) Explain how epigenetic changes could ensure that the 'gene output' from the X chromosomes in females is the same as in males (paragraph 19).

(2)

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(i) Explain why elderly men who have lost their Y chromosome have a higher risk of cancer (paragraphs 2, 3 and 20).

(3)

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